Math 252 [redacted] **Quiz 5** **Name:**

**1.** Evaluate $\int x\cos(4x)\,dx$

$u = x$, $dV = \cos(4x)\,dx$

$du = dx$, $V = \frac{1}{4}\sin(4x)$

$= \frac{1}{4}x\sin(4x) - \int \frac{1}{4}\sin(4x)\,dx$

$= \frac{1}{4}x\sin(4x) + \frac{1}{16}\cos(4x)\,dx + C$

**2.** Evaluate $\int \sin^3 x\cos^6 x\,dx$

$u = \cos x$

$du = -\sin x\,dx$

$= \int \sin^2 x\cos^6 x\ \sin x\,dx$

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$= \int \overbrace{(1-\cos^2 x)}^{(1-u^2)}\ \overbrace{\cos^6 x}^{u^6}\ \overbrace{\sin x\,dx}^{-du}$

$= -\int (1-u^2)u^6\,du$

$= -\int (u^6 - u^8)\,du$

$= -\left(\frac{1}{7}u^7 - \frac{1}{9}u^9\right) + C$

$= -\frac{1}{7}\cos^7 x + \frac{1}{9}\cos^9 x + C$

## Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$